

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

METAL TECHNOLOGY AND DESIGN 4055/1

PAPER 1 Theory, Drawing and Design

3 hours

Specimen paper

Additional materials:

Answer booklet,

Drawing paper A2 size (1 sheet),

Standard drawing equipment

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on all separate answer paper and used booklet.

Section A: Structured questions (40 marks)

Answer **all** questions.

Write your answers in the spaces provided.

Section B: Free Response Questions (30 marks)

Answer any **two** questions from this Section.

Write your answers on the separate answer booklets provided.

Section C: Compulsory question (30 marks).

Answer both questions **9** and **10**.

Answer these questions on the **A2** drawing paper provided.

At the end of the examination, fasten together your answer booklet to **Section A** and **B** and place them within your folded drawing paper for **Section C**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to spend no longer than **45 minutes** on **Section A**, no longer than **45 minutes** on **Section B** and **1 hour 30 minutes** on **Section C**.

Except where pictorial views are used, **all** diagrams are in **first** angle projection.

This specimen paper consists of 11 printed pages.

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SECTION A :28 MARKS

*Answer all questions in this section in the spaces provided.
You are advised not to spend more than 45 minutes.*

- 1 (a) Three possible cause of fire in the home are:
- (i) _____ [1]
- (ii) _____ [1]
- (iii) _____ [1]
- (b) Give any **two** examples of driving tools. [2]
- _____ [2]
- (c) Define *conductivity*.
- _____ [2]
- 2 (a) Give **one** safety precaution to be observed when cutting threads in a blind hole. [1]
- _____ [1]
- (b) State the difference between temporary joining and permanent joining methods. [2]
- _____ [2]
- (c) Calculate the overall length of rivet required to join two mild steel plates **6 mm** each using a diameter **5 mm** snap head rivet.

[4]

- 3 (a) The following abbreviations in relation to electricity stand for:

AC _____ [1]

DC _____ [1]

- (b) Give **two** reasons for applying surface finishes on metals. [2]

 _____ [2]

- (c) **Fig. 1** shows a defective flat chisel.

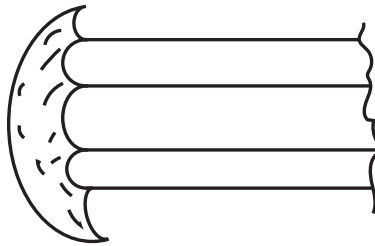


Fig.1

- (i) Name the defect shown.

_____ [1]

- (ii) Suggest a way of:

1. minimising the defect,

 _____ [1]

2. correcting the defect.

 _____ [1]

- 4 (a) List **two** advantages of using **CAD** drawing in design.

[2]

- (b) Identify **two** items made from plastics that are found in the workshop.

[2]

- (c) **Fig 2** shows a wheelbarrow, class 2 lever system.

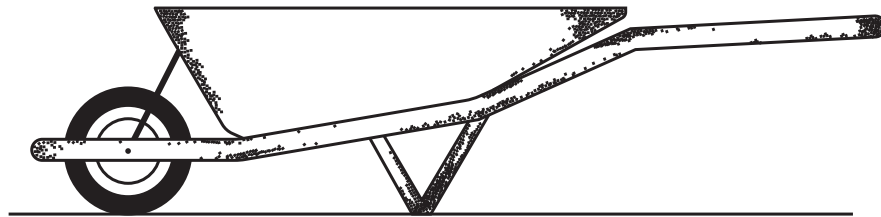


Fig .2

Label the fulcrum, load and effort.

[3]

Section B: 30 Marks

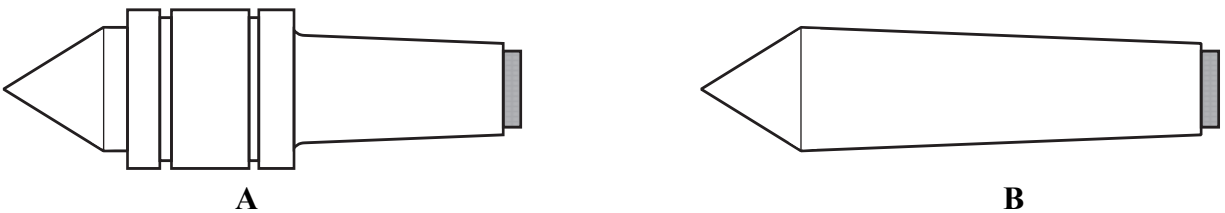
*Answer any **two** questions from this section.*

Use the separate answer sheets provided.

*You are advised to spend **not** more than 1 hour on this section.*

Option 1: Machines And Machining Processes

- 5 (a) State **three** injuries that can be caused by a running machine. [3]
- (b) (i) Name the **two** lathe centres shown in Fig 3. [3]

**Fig 3.**

- (ii) State the use of each of the centres. [2]
- (c) Use a sketch to show the process of boring on a lathe. [3]
- (d) State **two** operations that can be performed on a milling machine. [2]
- (e) Explain the use of the following parts of a milling machine:
- (i) main drive clutch, [1]
- (ii) table, [1]
- (iii) feed selector lever. [1]

Option 2: Welding Technology

- 6 (a) Name any **two** Personal Protective Equipment (PPE) to be worn to the body to protect it from flying hot particles during welding. [2]
- (b) **Figure 4** shows the welding positions during various welding operations. Name the welding positions marked (i) to (iv). [4]

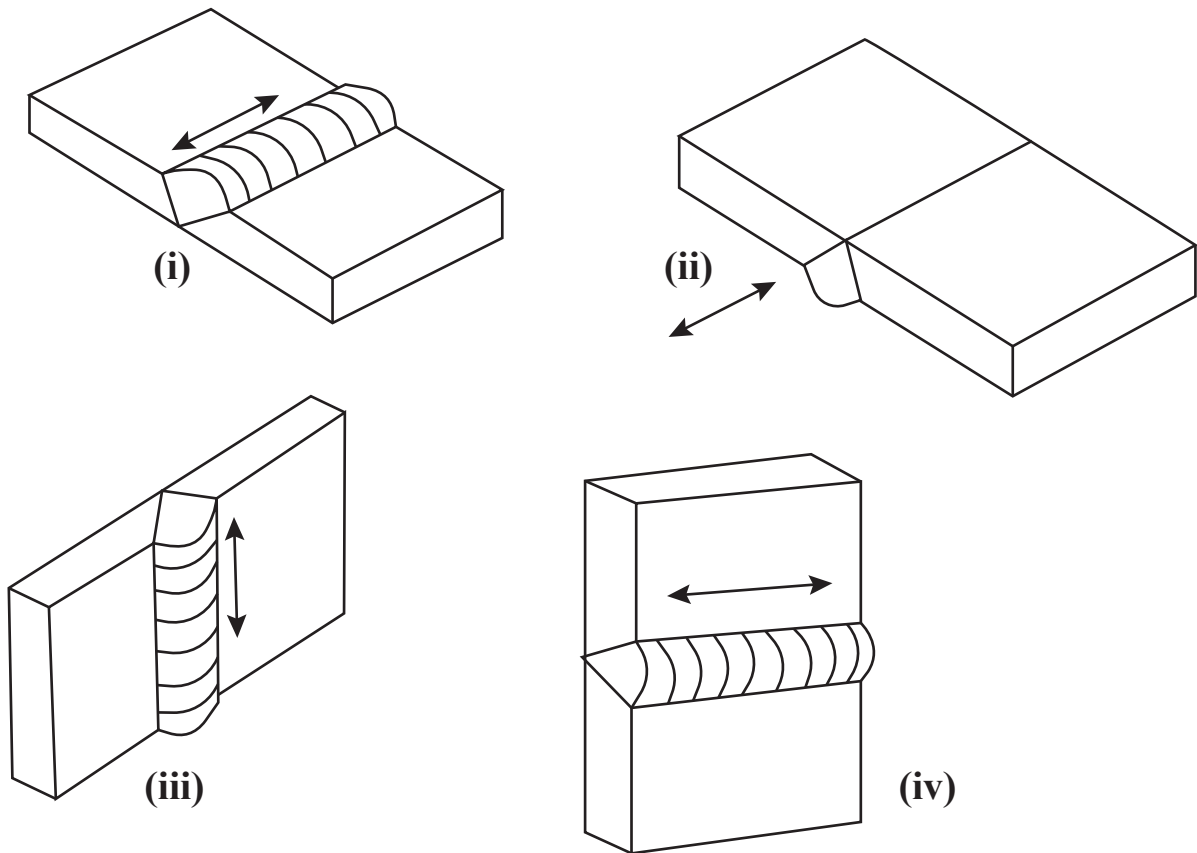


Fig.4

- (c) Differentiate between a gas cutting blowpipe and a welding blow pipe. [4]
- (d) Briefly explain the steps followed when shutting down the oxy-acetylene equipment.

OPTION 3: Sheet Metal And Beaten Metal Technology

- 7 (a) (i) Identify any **three** Sheet metal stakes shown in Fig. 5

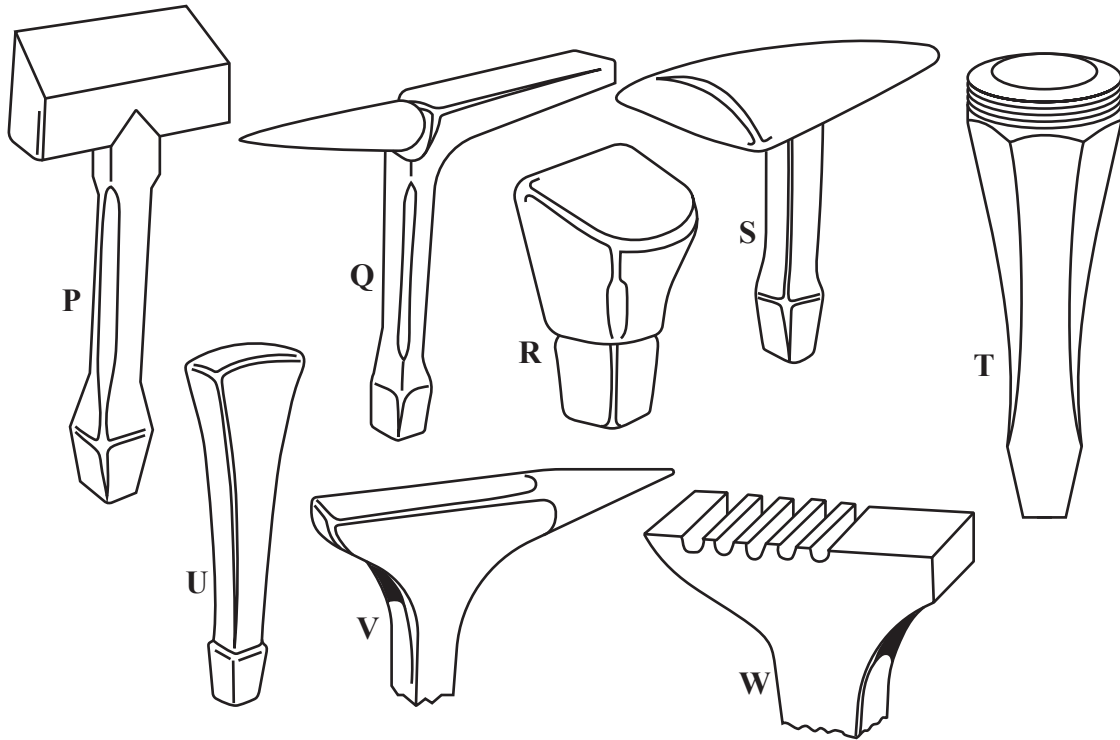


Fig.5

- (ii) Give the use of each of the stakes identified in (a) (ii). [3]
- (b) Explain **three** reasons why edge treatment is carried out on the edges of sheet metal articles. [3]
- (c) Define beaten metalwork. [1]
- (d) Identify any **two** items in the home, which are made using beaten metal technology. [2]
- (e) Explain with the aid of sketches, the process of raising in beaten metal technology. [3]

OPTION 4: Forge Technology And Foundry Technology

- 8** **(a)** Identify any **two** hazards associated with forge technology. [2]
- (b)** Give the use of the following tools in forge technology:
- (i)** chisels and [1]
- (ii)** tongs. [1]
- (c)** Design an item that can be used to hang flowers in the home.
 The item should include at least three forge technology processes. [6]
- (d)** State the difference between sand casting and die casting. [1]
- (e)** **(i)** Explain the following casting defects
- porosity [1]
- inclusion [1]
- (ii)** Outline the possible cause of each of the defects. [2]

SECTION C: (22 MARKS)

Answer all the questions in this section on the A2 drawing paper provided.

*You are advised to spend **1 hour 30 minutes** on this section. One sheet of A2 drawing paper to be used. Use **one** side of the paper. Set the paper with the long edge to the top of your drawing board. Use the space to the right of the vertical line to make sketches in answer to Question 9 (a) and b.*

- 9** Fig 9 shows an incomplete design of a Light Duty Press Stand.

In its complete form:

Part C should be locked at any point along the column **Part B**;

Part D should slide up and down through **Part C** as the press is fed or lifted. **Part D** should be fitted with a quick release mechanism. (see Fig 9)

Design:

- (a) A method of safety locking **Part C** at any point along the column **Part B**. [5]
- (b) A quick release mechanism for allowing the press to return to the starting point once the operation is completed. [5]

- 10** Draw to a scale of **1:2** in First Angle projection the following views of the fully Assembled Light Duty Press stand in Fig 9 including your solutions to the design problems.

- (a) Front elevation in the direction of arrow **F**. With the centre of **Part C** **300 mm** above the top of the table as shown on Fig.9. [5]
- (b) A sectional end elevation taken along **Part D** to include details of quick releasing mechanism (**Section E – E**) in Fig 10. [5]
- (c) Add **two** main dimensions to your drawing. [2]

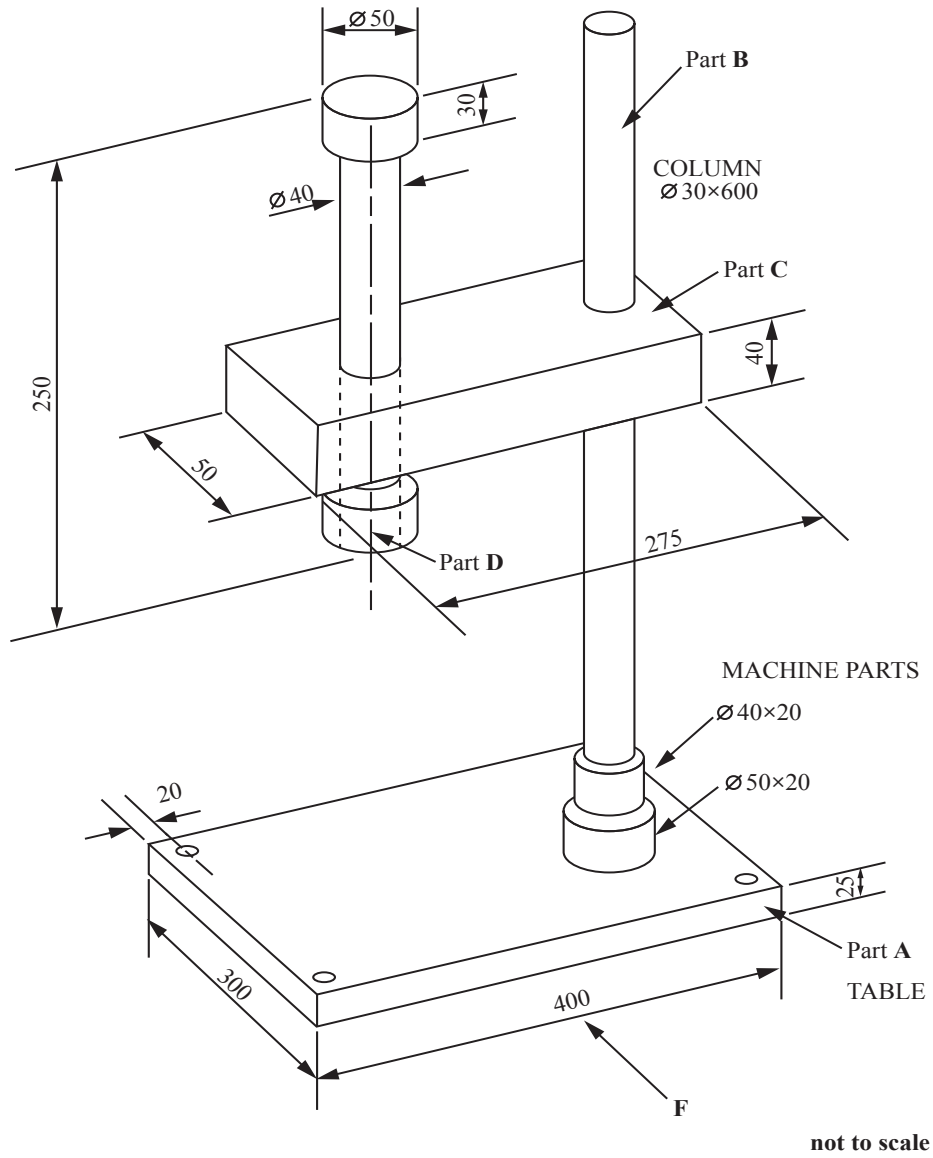


Fig 9 Light Duty Press Stand

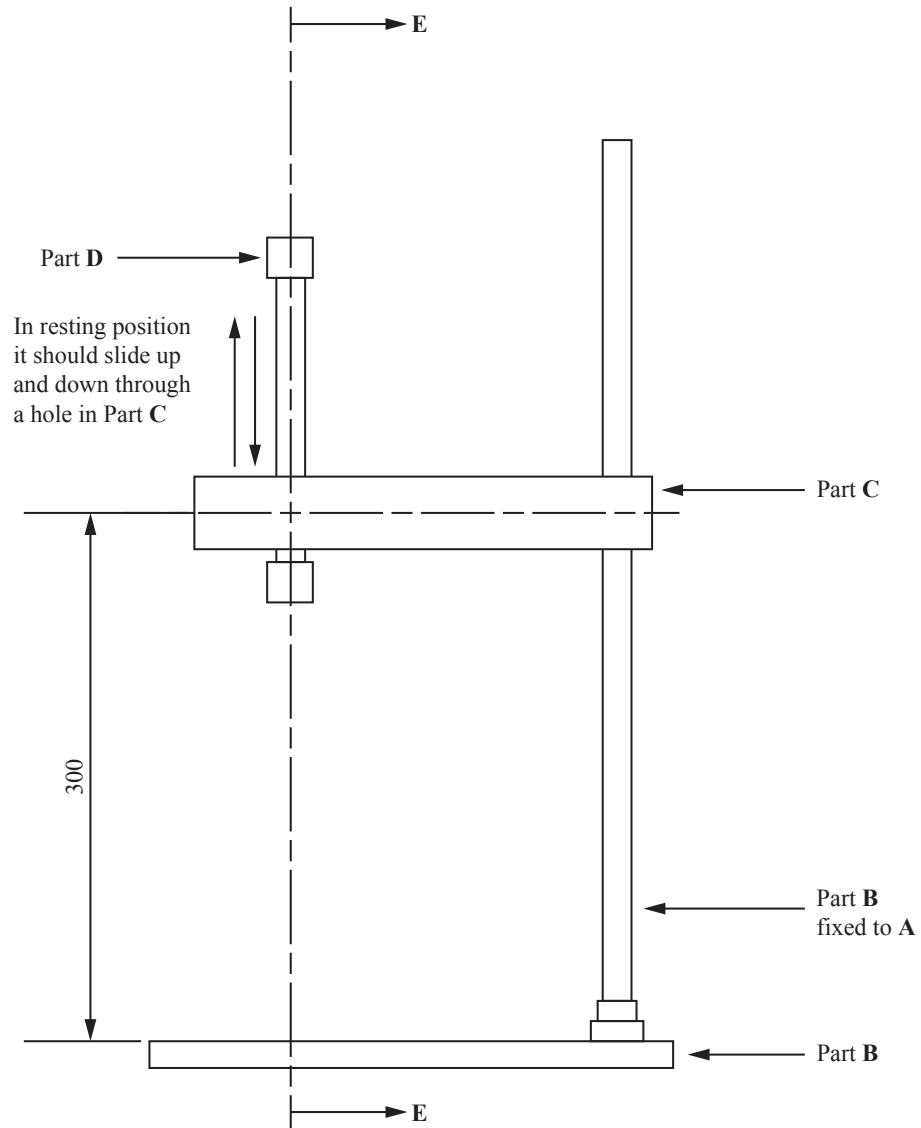


Fig 10 Light Duty Press Stand